

TOPIC 11/60

What are Parameters?

The "neurons" of an LLM.

How AI stores knowledge.

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THE "KNOBS"

Think of an LLM as a giant machine with billions of tiny dials (knobs).

What they are:

Parameters are just numerical values (weights and biases) that the model adjusts during training to learn patterns.

What they do:

They control how much weight a piece of information gets when the model is processing text.

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THE ANALOGY



The Sound Mixer

Imagine a professional sound mixer with 1 trillion channels.

To create the perfect sound, you have to tune every single one of those 1 trillion dials perfectly. Training an LLM is just the act of finding the perfect settings for every dial.

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KEY MISCONCEPTION

Not a Database

They aren't facts.

A common mistake: thinking an LLM "stores facts" like a database. It doesn't.

Parameters store patterns.

They store the *relationship* between words, grammar, logic, and facts. When you ask a question, the model reconstructs the answer based on these patterns.

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THE GROWTH

Scale

Bigger is better (usually)

As we increase parameters, the model gets exponentially smarter at reasoning, coding, and nuance.

GPT-2: 1.5 Billion

GPT-3: 175 Billion

GPT-4: Estimated 1 Trillion+

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THE REAL-WORLD COST

Compute

Memory Matters

Parameters take up physical space (VRAM) on GPUs.

Each parameter typically takes 2 bytes (or more) of VRAM. A 70-billion parameter model needs ~140GB of VRAM just to *run*, which is why your local laptop can't run the biggest models.

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SCALING LAWS

THE RULE

To keep getting smarter, you need more data AND more parameters.

> If you increase params but not data, you get "overfitting".

THE EFFICIENCY SHIFT

Small, dense models are the new trend.

We are now learning to build 7B parameter models that outperform 175B models thanks to better training data quality (e.g., Llama 3).

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FINE-TUNING INNOVATION

LoRA

How do you train a 1-trillion parameter model on a budget?
You don't tune all of them.

LoRA (Low-Rank Adaptation) lets you freeze the original model and add tiny "adapters" on top. Instead of retraining 1 Trillion parameters, you only tune a few Million.

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THE BIG PICTURE

Complexity

Parameters are the storage of the AI's "world view".

Hardware Bottleneck

Hardware is literally the limit. We need better chips to run more parameters.

Efficiency Revolution

We are moving from "make it bigger" to "make it smarter".

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Parameters

CONCEPT UNLOCKED



Synaptic Knobs



Store Patterns, not Facts



Bigger = Generally Smarter



Efficient tuning = LoRA

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